Learning Healthy Anatomy to Find Tumors: Comparing VAEs and GANs for Unsupervised Anomaly Detection in Brain MRI (Midway Report)

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Abstract

Supervised tumor segmentation needs thousands of expert-drawn outlines and only recognizes the diseases it was trained on. We instead train generative models only on *healthy* brain MRI and flag the regions a model cannot reconstruct. We compare a variational autoencoder (VAE), f-AnoGAN (a Wasserstein GAN with an encoder) and a plain autoencoder (AE) under one shared pipeline. Models are trained on 462 healthy IXI subjects (T2) and tested on 436 glioma patients from the Medical Segmentation Decathlon. So far we have built the full pipeline, including our own PyTorch affine registration, and evaluated all three models at 64×64. In preliminary single-seed runs the VAE is best (subject Dice 0.51, pixel AUROC 0.89), f-AnoGAN is close (Dice 0.47) with the fewest complete misses, and the plain AE does not beat a simple "bright on T2" intensity baseline on Dice (0.38 vs. 0.39). Our main challenges were registration failures and CPU-only training. Next we will move to the proposal's 128×128 resolution, add seeds and confidence intervals, and sweep the VAE's KL weight and latent size.

1. Introduction

Most deep learning tools for medical imaging are supervised: they learn from scans in which a radiologist has outlined every lesion. Such labels are slow and expensive to produce, and a supervised model only recognizes the abnormalities it was shown. Healthy scans, by contrast, are plentiful.

We ask whether a generative model that has only ever seen healthy brains can detect and localize brain tumors, and whether a VAE or a GAN does this better. The idea is to learn the distribution of healthy anatomy, $p(\mathbf{x})$. When the model is shown a scan with a tumor, it "repairs" the image toward the nearest healthy-looking brain $\hat{\mathbf{x}}$, and the residual $\mathbf{x} - \hat{\mathbf{x}}$ highlights what does not belong. Because no tumor labels are used for training, this *unsupervised anomaly detection* could flag abnormalities nobody trained it to find.

Generative models are central to this approach. A plain autoencoder only learns to copy its input, and with enough capacity it copies tumors too. A VAE (latent prior) and a GAN (realism critic) both restrict reconstructions to images that look like the training distribution. VAEs are stable but blurry, so fine edges cause false alarms. GANs are sharp but harder to train and may not cover all of healthy anatomy. Published comparisons disagree, partly because they use different private data and preprocessing (Baur et al., 2021). Our contribution is a controlled comparison on public data, with identical preprocessing and post-processing for every model.

2. Related Work

Generative models. VAEs (Kingma & Welling, 2014) maximize a lower bound on the data likelihood with an encoder–decoder pair and a Gaussian latent prior. GANs (Goodfellow et al., 2014) train a generator against a discriminator; DCGAN (Radford et al., 2016) and WGAN-GP (Gulrajani et al., 2017) made this stable on images. We use DCGAN-style layers for all models so that differences come from the training objective, not the architecture.

GANs for anomaly detection. AnoGAN (Schlegl et al., 2017) was the first to use a GAN trained on healthy images to localize anomalies in medical images. It finds the closest healthy image by optimizing the latent code separately for every test image, which is slow. f-AnoGAN (Schlegl et al., 2019) replaces this search with an encoder trained after the WGAN-GP, giving a single forward pass. We adopt f-AnoGAN as our GAN representative.

Autoencoders on brain MRI. Baur et al. (2018) combined a VAE with an adversarial loss to segment multiple-sclerosis lesions without labels. Chen & Konukoglu (2018) constrained the latent space of an adversarial autoencoder for

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STAT 453 Midway Report, November 2026.

brain lesions, and Zimmerer et al. (2019) showed that VAE gradients and the KL term also carry localization information. The comparative study of Baur et al. (2021) benchmarked many AE and GAN variants on shared data. It found that results depend heavily on preprocessing and postprocessing, and that no model family clearly wins. Meissen et al. (2021) went further: on FLAIR scans a simple intensity threshold matched or beat these deep models, because tumors are simply hyperintense. More recent work uses diffusion models (Pinaya et al., 2022), but VAEs and GANs remain the standard reference points.

Our position. We therefore use only public data (Imperial College London; Antonelli et al., 2022; Menze et al., 2015), share all preprocessing and scoring between models, and add a plain AE and the intensity baseline of Meissen et al. (2021) as controls.

3. Methods (Implemented)

3.1. Preprocessing pipeline

IXI and MSD come from different scanners and were processed differently, so a model could flag "different hospital" instead of "tumor". We therefore apply one pipeline to every scan: (1) reorient to RAS and roughly rescale intensities; (2) affinely register (12 parameters) to the MNI152 2009a template (Fonov et al., 2011); (3) skull-strip with the template's brain mask (MSD scans are already skull-stripped, so we also intersect with their own brain outline); (4) divide by the 99th percentile of brain intensities and clip to $[0, 1]$; (5) keep axial slices whose brain area is at least 30% of the maximum, taking every second level, crop around the brain and resize to 128×128 by area averaging. The 64×64 inputs are pooled from these slices.

Challenge 1: no registration library. Our team develops on Windows laptops with ARM processors (Snapdragon X Elite). SimpleITK, ANTs and dipy have no builds for this platform, and neither does any dedicated skull-stripping tool. We therefore wrote affine registration ourselves in PyTorch. It resamples with `grid_sample`, maximizes normalized cross-correlation (NCC) with Adam, works coarse to fine (4 mm, then 2 mm), and uses a random subset of 40k voxels per step for speed. A unit test confirms it recovers a known affine transform to under 1 mm. Nilearn ships only a T1 template, so we built a T2-like template from its gray- and white-matter probability maps (CSF $= 1.0$, GM $= 0.55$, WM $= 0.35$, matching T2 contrast).

Challenge 2: registration failures. In our first full run, about 30% of IXI scans registered badly (NCC ≈ 0, with the eyes inside the brain mask). Head tilt and a poor centre-of-mass start trapped the optimizer in local minima. What worked was a coarse 6 mm grid search over pitch ($\pm 25°$), shifts ($\pm 20/\pm 40$ mm) and scale (0.8–1.0), refining the three best starts. All 578 IXI and 484 MSD scans then pass a visual check (median NCC 0.47 and 0.25); the lowest MSD scores are well-aligned scans with very large tumors.

3.2. Models

All models share DCGAN-style networks. The encoder applies four stride-2 4×4 convolutions ($1 \to 32 \to 64 \to 128 \to 256$ channels, LeakyReLU 0.2, batch norm), turning a 64×64 image into a $256 \times 4 \times 4$ feature map, followed by a linear layer. The decoder mirrors this with transposed convolutions and a sigmoid output. The latent size is $d = 128$, and weights follow the DCGAN initialization.

AE (2.44M parameters) minimizes $\|\mathbf{x} - D(E(\mathbf{x}))\|_2^2$.

VAE (2.96M) minimizes the negative ELBO with a Gaussian decoder:

$$\mathcal{L}_{\text{VAE}} = \|\mathbf{x} - D(\mathbf{z})\|_2^2 + \beta \,\mathrm{KL}\big(q(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(\mathbf{0}, I)\big), \quad (1)$$

with $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$ (the reparameterization trick) and $\beta = 1$. At test time we decode the posterior mean.

f-AnoGAN (3.13M) is trained in two stages. Stage 1 is a WGAN-GP: the generator G is the decoder architecture and the critic C is the encoder architecture with layer normalization instead of batch normalization, because the gradient penalty is computed per sample. The critic minimizes $\mathbb{E}[C(G(\mathbf{z}))] - \mathbb{E}[C(\mathbf{x})] + \lambda\, \mathbb{E}[(\|\nabla_{\tilde{\mathbf{x}}} C(\tilde{\mathbf{x}})\|_2 - 1)^2]$ with $\lambda = 10$ and $n_{\text{critic}} = 5$. Stage 2 freezes G and C and trains an encoder E with the izi$_f$ loss $\|\mathbf{x} - G(E(\mathbf{x}))\|^2 + \kappa \|f(\mathbf{x}) - f(G(E(\mathbf{x})))\|^2$, where f denotes the critic's last feature map and $\kappa = 1$.

3.3. Anomaly scoring

Following (A3), our main anomaly map keeps only the *positive* residual:

$$A = \big(g_\sigma * \max(\mathbf{x} - \hat{\mathbf{x}},\, 0)\big) \odot M_\ominus, \quad (2)$$

where g_σ is a Gaussian ($\sigma = 1$ pixel) and $M_\ominus$ is the brain mask eroded by 2 pixels, because reconstructions are least reliable at the brain border. A pixel is labelled tumor if $A > \tau$. The threshold τ maximizes Dice on a held-out MSD validation set. A slice's score is the mean of its top 1% of anomaly values. We also report the plain absolute residual $|\mathbf{x} - \hat{\mathbf{x}}|$ as an ablation. The *intensity baseline* uses $\hat{\mathbf{x}} = \mathbf{0}$, so it detects "bright = tumor".

3.4. Assumptions

We state the method's assumptions explicitly, since each can fail. **(A1) Healthy training data:** every IXI scan is free of pathology (IXI volunteers were not screened for incidental findings). **(A2) Anomalies are not reconstructed:**

a model trained only on healthy anatomy maps a tumor to healthy tissue. This requires that it cannot simply copy its input; the AE shows this can fail (Section 4.2). **(A3) Tumors are bright on T2:** edema and most tumor tissue are hyperintense, so we score only $\mathbf{x} - \hat{\mathbf{x}} > 0$; dark parts such as necrotic cores are missed. **(A4) Comparable inputs:** after affine registration, one shared template mask and 99th-percentile scaling, the residual reflects pathology rather than scanner or site differences between IXI and MSD. Affine alignment does not match individual folds, and the mask is approximate at the brain border. **(A5) Independent 2D slices:** slices are scored without 3D context, and left–right flips assume healthy brains are roughly symmetric. **(A6) Labels only for the threshold:** the merged whole-tumor mask is the reference, and one global τ, chosen on 48 labelled MSD subjects assumed to match the test distribution, is used for all subjects. **(A7) VAE likelihood:** the summed squared error corresponds to a Gaussian decoder with the same fixed variance for every pixel.

Challenge 3: compute. All results so far were run without a GPU, on a Windows laptop with an ARM processor. The x64 PyTorch build under emulation (oneDNN kernels) trained 2.6× faster than the native ARM64 build (670 vs. 257 images/s), so we use it, with the three models in parallel. This limit is why we use the proposal's 64×64 "fast experiment" setting.

4. Experiments

4.1. Data and setup

Training data: IXI (Imperial College London), 578 healthy adults, T2-weighted, split 80/10/10 by subject into 462 training, 58 validation and 58 test subjects (26,796 training slices). **Test data:** MSD Task01 BrainTumour (Antonelli et al., 2022), 484 glioma patients from BraTS (Menze et al., 2015). We use the T2 channel and merge all tumor labels into one "whole tumor" mask. 48 subjects (10%) are used only to choose the threshold τ, and 436 subjects (25,288 slices) are the test set. Tumor pixels make up 9.2% of MSD test brain pixels. **Training:** Adam with learning rate $2{\cdot}10^{-4}$ and batch size 64, random left–right flips, 40 epochs for AE and VAE. f-AnoGAN uses 60 WGAN-GP epochs (learning rate 10^{-4}, $\beta = (0, 0.9)$) followed by 30 encoder epochs. We keep the checkpoint with the lowest healthy validation MSE. Training took 2.9 h for AE and VAE and 6.3 h for f-AnoGAN on CPU. **Metrics** follow the proposal: Dice, pixel AUROC and AUPRC, two slice AUROCs, and the false-positive rate on healthy IXI test pixels.

4.2. Preliminary results

Training behaved as expected (Figure 2). The AE reaches by far the lowest healthy reconstruction error ($2.1 \cdot 10^{-3}$, vs. $4.7 \cdot 10^{-3}$ for the VAE and $5.6 \cdot 10^{-3}$ for f-AnoGAN). The Wasserstein estimate falls steadily from 35.5 to 1.5 without collapse.

The VAE and f-AnoGAN outperform the AE and intensity baseline on subject-level Dice in our preliminary experiments (Table 1). The VAE is best on every pixel-level metric. It improves subject Dice by about 0.13 over the AE and 0.12 over the intensity baseline, while reducing healthy-pixel false positives by factors of 3.3 and 3.7, respectively. f-AnoGAN comes second on pixel metrics and ties the VAE at slice level (AUROC 0.93 vs. healthy IXI). The plain AE, despite reconstructing healthy brains best, has the *lowest* pixel AUROC (0.80). Figure 1 shows why: it reconstructs part of the tumor too, so the residual misses it, violating (A2): good reconstruction is not the goal; reconstructing *only* healthy anatomy is. The intensity baseline is strong at pixel level (AUROC 0.85), in line with Meissen et al. (2021), but poor at slice level (0.77), since many bright healthy structures such as ventricles and CSF also trigger it.

Post-processing matters. With the absolute residual instead of the positive one, every model loses Dice (VAE $0.514 \rightarrow 0.484$, f-AnoGAN $0.474 \rightarrow 0.435$, AE $0.380 \rightarrow 0.334$). The healthy FPR of f-AnoGAN doubles ($1.3\% \rightarrow 2.6\%$), because dark mismatches at fine folds are now counted.

Small tumors are the main failure mode (Figure 3). Splitting test subjects into thirds by tumor size, VAE Dice rises from 0.34 (small) to 0.65 (large), and every model shows this pattern. The VAE leads on large tumors (0.65 vs. 0.57), but f-AnoGAN has the fewest near-complete misses: 8.7% of subjects have Dice < 0.2, compared with 11.7% for the VAE, 14.7% for the AE and 28.2% for the baseline. f-AnoGAN's scattered false positives thus mostly hurt on large tumors, where Dice depends on boundary quality.

Caveats. Each number comes from a single training run (seed 42), so differences of a few Dice points are not yet established. Assumption (A4) holds only partly: template-mask skull stripping leaves a thin rim of skull on some IXI scans, so part of every residual reflects domain shift.

5. Conclusion and Updated Plan

Progress. In our single-seed 64×64 runs, both generative models beat a plain AE, the VAE is best on average and f-AnoGAN has the fewest near-complete misses. **Changes from the proposal:** our own registration and T2 template (no standard tool runs on our hardware), plus the intensity baseline and positive-residual scoring from Meissen et al. (2021).

Next steps (by priority). (1) *Robustness:* 3 seeds per model, bootstrap confidence intervals over subjects, and a paired

Table 1. Test results at 64×64 (positive residual). Dice is computed per subject (mean $\pm$ sd over 436 MSD test subjects) and over all test pixels pooled (global). Slice AUROC (MSD) separates tumor slices from tumor-free slices of MSD patients. Slice AUROC (vs. IXI) separates MSD tumor slices from healthy IXI test slices. FPR is the fraction of healthy IXI test brain pixels flagged at the chosen threshold. Best value per column in bold.

Model	Dice (subject)	Dice (global)	Pixel AUROC	Pixel AUPRC	Slice AUROC (MSD)	Slice AUROC (vs. IXI)	Healthy FPR
Intensity baseline	0.394 ± 0.243	0.431	0.854	0.351	0.779	0.772	4.52%
AE	0.380 ± 0.161	0.420	0.796	0.402	**0.869**	0.865	4.01%
VAE	$\mathbf{0.514 \pm 0.227}$	**0.566**	**0.894**	**0.575**	0.867	0.929	**1.21%**
f-AnoGAN	0.474 ± 0.183	0.517	0.855	0.514	0.867	**0.931**	1.31%

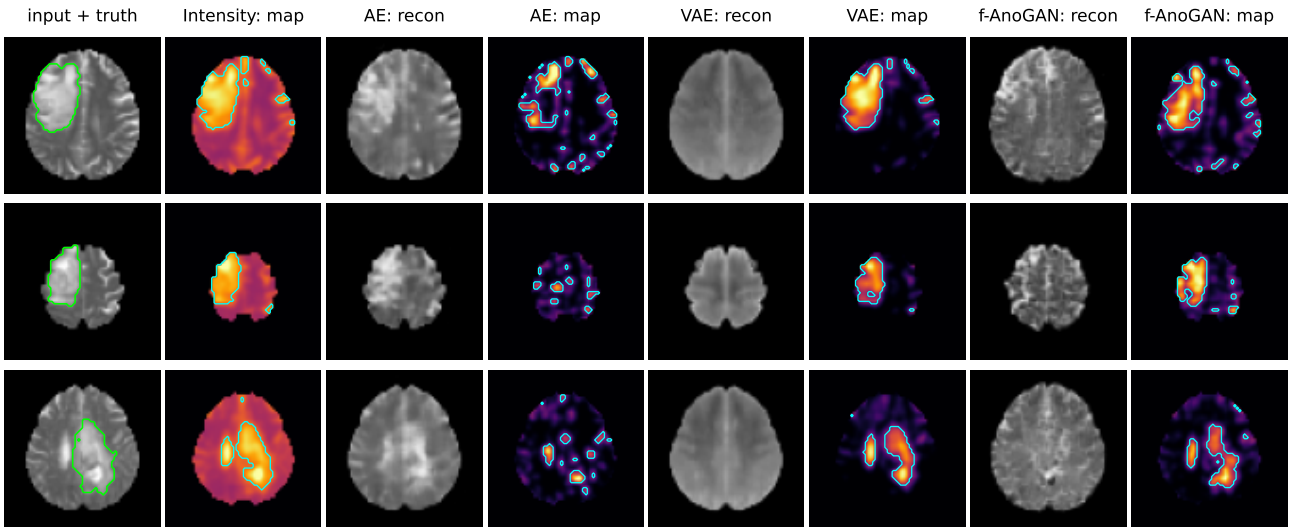


Figure 1. MSD test slices (rows) with each model's healthy reconstruction and anomaly map. Green: expert whole-tumor outline. Cyan: predicted tumor at the threshold chosen on validation. The VAE's blurry reconstruction leaves one compact anomaly. f-AnoGAN is sharper but adds scattered false positives where folds are misplaced. The AE partly reconstructs the tumor itself, which weakens its map.

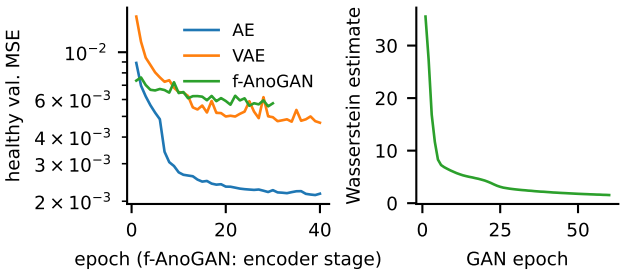


Figure 2. Left: reconstruction error on healthy IXI validation slices. Right: the f-AnoGAN critic's Wasserstein estimate during GAN training.

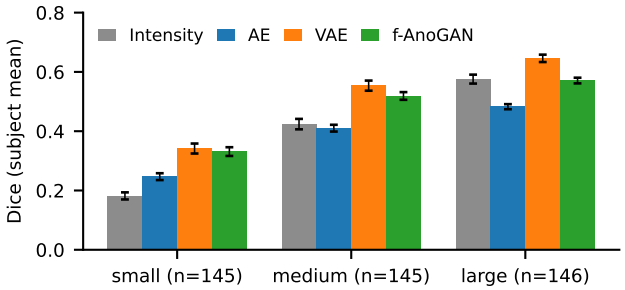


Figure 3. Subject Dice by tumor-size tertile (whole-tumor pixels summed over a subject's 64×64 slices). Error bars show the standard error.

Wilcoxon test of VAE vs. f-AnoGAN Dice. (2) *Resolution:* train at 128×128, the proposal's main setting. This needs about $4\times$ more compute (over 24 h for f-AnoGAN on the laptop CPU), so we plan to migrate training to a Google Colab GPU runtime. (3) *VAE trade-off:* sweep $\beta \in \{0.5, 1, 2\}$ and latent size $d \in \{128, 256\}$ to measure how blur trades sharpness against false alarms. (4) *Error analysis:* results by tumor sub-region, and whether HD-BET (Isensee et al., 2019) skull stripping on Colab reduces border false positives (testing A4).

Timeline. *Nov 9–20:* seed repeats and confidence intervals; migrate to a Colab GPU runtime; start 128×128 runs. *Nov 23:* β/latent sweep; sub-region analysis. *Nov 30:* HD-BET ablation; freeze experiments. *Dec 7:* final report.

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